MOHAMMED ROSHAN

B.Tech Information Technology Graduate (2026)

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Nationality: Indian | GitHub: github.com/Mohammed7025

PROFILE

B.Tech Information Technology graduate (2026) with hands-on experience building AI-enabled full-stack and IoT systems, from sensor hardware and machine-learning models to backend APIs and web and mobile interfaces. Seeking a Master's degree in Information Technology, Data Science or Artificial Intelligence to deepen expertise in machine learning and data-driven systems.

Areas of Interest: Applied Machine Learning, Explainable AI, IoT-Enabled Intelligent Systems, Natural Language and Sentiment Analysis

EDUCATION

Qualification	Institution	Board / University	Years	Result
Bachelor of Technology (B.Tech), Information Technology	MEA Engineering College, Perinthalmanna, Kerala, India	APJ Abdul Kalam Technological University	2022 – 2026	**CGPA 7.87/10**
Higher Secondary Education (Computer Science)	GMHSS Perinthalmanna, Kerala, India	Board of Higher Secondary Examinations, Government of Kerala	2020 – 2022	**91%**
Secondary Education (Class 10)	PHSS Pandallur, Kerala, India	General Education Department, Government of Kerala	2019 – 2020	**99%**

RELEVANT COURSEWORK

Subject	Semester	Grade
Vector Calculus, Differential Equations and Transforms	2	**A+**
Data Structures	3	**A**
Algorithm Analysis and Design	6	**A+**
Data Science	6	**A**
Soft Computing	6	**A**
Machine Learning	7	**A+**
Data Analytics	7	**A+**
Computer Graphics	8	**A+**
Internet of Things	8	**A+**
Software Architecture	8	**A**

PROJECTS

AI-Driven Smart Agriculture System | B.Tech Major Project (Project Lead, Team of 5) **2025 – 2026**

Guide: Ms. Thasleena V A, Head of Department, Information Technology, MEA Engineering College

Problem and Approach

- Small and medium farmers often lack affordable soil testing and expert advice. Built an end-to-end, low-cost platform that reads live soil and weather data and returns crop, fertilizer and irrigation recommendations with explanations a farmer can follow.
- Worked across all layers of the system: sensor hardware, ML models, backend API, and web and mobile interfaces.

Hardware and Data Pipeline

- NodeMCU (ESP8266) reading NPK, moisture, pH, temperature and humidity sensors, powered through a 12V to 3.3V regulator, and readings sync to Firebase in real time.
- Combined live soil readings from the sensors, real-time weather data fetched by location, and soil type as inputs to the models.

Machine Learning

- Trained Transformer-based models on publicly available datasets to recommend crops and fertilizers and to decide when irrigation should be on or off, weighting the input features differently for each task.

- Added SHAP-based explainability, so each recommendation shows which factors influenced the result.

Software

- FastAPI backend with JWT authentication, a React web dashboard, a mobile app with a Malayalam language option, and AgroBot, a chatbot for crop, soil, irrigation and fertilizer questions.

Tech: Python, PyTorch, TensorFlow, scikit-learn, FastAPI, React, Firebase, NodeMCU (ESP8266), SHAP | Code: github.com/Mohammed7025/Smart-Agri-using-AI-and-iot

AI Dietitian and Mental Health Tracker | B.Tech Mini Project (Project Lead, Team of 5) **Apr 2025**

Guide: Ms. Thasleena V A, Head of Department, Information Technology, MEA Engineering College

Problem and Approach

- Led a team of five in developing a full-stack web platform that combines personalised diet planning with mental-wellness tracking.

Key Features

- Rule-based diet planner using BMI, dietary preferences and medical conditions, with a decision-tree fallback for edge cases.
- Integrated the Google Gemini API for a supportive chatbot and for sentiment analysis of journal entries.
- Daily mood, stress and sleep logging with interactive trend charts, and secure sign-in with Firebase Authentication.
- Evaluated through user feedback.

Tech: Node.js, MongoDB Atlas, Firebase Authentication, JavaScript, HTML, CSS, Gemini API | Code: github.com/Mohammed7025/Ai-dietitian-and-mental-health-tracker

TECHNICAL SKILLS

Languages: Python, JavaScript

Machine Learning and Data: scikit-learn, PyTorch, TensorFlow, Pandas, NumPy, SHAP, Matplotlib, Plotly

Backend and APIs: FastAPI, Node.js, REST API Development, JWT Authentication, Postman, n8n Workflow Automation, Gemini API Integration

Databases: MongoDB Atlas, Firebase, MySQL, PostgreSQL (Supabase)

Frontend and IoT: React, HTML, CSS, NodeMCU (ESP8266), Sensor Interfacing, Arduino IDE

Tools: Git, GitHub, VS Code

CERTIFICATIONS

- **Python for Data Science** – NPTEL (IIT Madras, SWAYAM), Elite + Silver, Jul–Aug 2025
- **Python: Introduction to Data Science and Machine Learning A-Z** – Udemy, Apr 2025
- **AI for Entrepreneurship** – Intel, Skill India Digital Hub, Jun 2025

ENGLISH PROFICIENCY (IELTS)

Listening	Reading	Writing	Speaking	Overall	Test Date	TRF No.
8.5	9.0	7.0	7.5	**8.0**	28 Sep 2026	26IA525837M008A

OTHER LANGUAGES

Language	Proficiency Level
Malayalam	Native
Hindi	Advanced
Arabic	Intermediate
Urdu	Intermediate

LEADERSHIP, WORKSHOPS AND TRAINING

- **MEA FOSS** – Active member, Free and Open Source Software initiatives, MEA Engineering College
- **XORIT (IT Association)** – Assistant Technical Coordinator
- **Young Innovators Programme 8.0** – Voice of Stakeholder training, Kerala Development and Innovation Strategic Council (K-DISC), Dec 2025
- **Basic Life Support (BLS) Course** – Aster MIMS, Calicut (American Heart Association affiliated), Aug 2025
- **iOS App Development Workshop** – iPlanet Education (Apple Authorised Education Specialist), Dec 2022